

[BUG] Incorrect Expectation Values of Scalar Product of Sum which is non

<https://github.com/PennyLaneAI/pennylane/issues/3198>

ROW 26

[BUG] Incorrect Expectation Values of Scalar Product of Sum which is non-hermitian #3198

New issue

Closed



Jaybsoni opened on Oct 21, 2022

Expected behavior

The results of these two expectation values should be the same.

```
import pennylane as qml
from pennylane import numpy as np

dev = qml.device("default.qubit", wires=1)

@qml.qnode(dev)
def circuit(op):
    # |psi_0>: state preparation
    qml.RX(np.pi / 7, wires=0)
    qml.RY(np.pi / 3, wires=0)
    qml.RZ(np.pi / 7, wires=0)

    # Gives the expectation value relative to input operator
    return qml.expval(op)

# Define the operator sigma_x sigma_y
op = qml.prod(qml.PauliX(0), qml.adjoint( qml.PauliY(0)) )
# Define its adjoint adj(sigma_x sigma_y)
op_adj = qml.adjoint(op)
Imag = -0.5j * (op - op_adj)

assert np.allclose(qml.matrix(Imag), qml.matrix(qml.PauliZ(0)))

print(circuit(qml.PauliZ(0)))
print(circuit(Imag))
```

Actual behavior

The results are different, a casting error occurs when computing the Eigen decomposition and the complex Eigen values of the sum operator are lost.

Additional information

No response

Source code

No response

Tracebacks

No response

System information

```
Name: PennyLane
Version: 0.27.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/XanaduAI/pennylane
Author:
Author-email:
License: Apache License 2.0
Location: /opt/anaconda3/envs/pl_env/lib/python3.9/site-packages
Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-lightning, retworkx, scipy, semar
Required-by: amazon-braket-pennylane-plugin, PennyLane-Forest, PennyLane-Lightning, PennyLane-qiskit

Platform info: macOS-10.16-x86_64-i386-64bit
```

Assignees

- AlbertMitjans
- Jaybsoni
- co9olguy

Labels

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications

Customize

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Participants



Visual Studio

Name: PennyLane
Version: 0.27.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: <https://github.com/XanaduAI/pennylane>
Author:
Author-email:
License: Apache License 2.0
Location: /opt/anaconda3/envs/pl_env/lib/python3.9/site-packages
Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-lightning, retworkx, scipy, semar
Required-by: amazon-braket-pennylane-plugin, PennyLane-Forest, PennyLane-Lightning, PennyLane-qiskit

Platform info: macOS-10.16-x86_64-i386-64bit
Python version: 3.9.13
Numpy version: 1.23.2
Scipy version: 1.9.1
Installed devices:
- default.gaussian (PennyLane-0.27.0.dev0)
- default.mixed (PennyLane-0.27.0.dev0)
- default.qubit (PennyLane-0.27.0.dev0)
- default.qubit.autograd (PennyLane-0.27.0.dev0)
- default.qubit.jax (PennyLane-0.27.0.dev0)
- default.qubit.tf (PennyLane-0.27.0.dev0)
- default.qubit.torch (PennyLane-0.27.0.dev0)
- default.qutrit (PennyLane-0.27.0.dev0)
- braket.aws.qubit (amazon-braket-pennylane-plugin-1.7.0)
- braket.local.qubit (amazon-braket-pennylane-plugin-1.7.0)
- lightning.qubit (PennyLane-Lightning-0.26.0)
- forest.numpy_wavefunction (PennyLane-Forest-0.25.0.dev0)
- forest.qvm (PennyLane-Forest-0.25.0.dev0)
- forest.wavefunction (PennyLane-Forest-0.25.0.dev0)
- qiskit.aer (PennyLane-qiskit-0.25.0.dev0)
- qiskit.basicaer (PennyLane-qiskit-0.25.0.dev0)
- qiskit.ibmq (PennyLane-qiskit-0.25.0.dev0)
- qiskit.ibmq.circuit_runner (PennyLane-qiskit-0.25.0.dev0)
- qiskit.ibmq.sampler (PennyLane-qiskit-0.25.0.dev0)

Existing GitHub issues

☒ I have searched existing GitHub issues to make sure the issue does not already exist.



Jaybsoni added **bug** on Oct 21, 2022



Jaybsoni assigned [co9olguy](#), [Jaybsoni](#) and [AlbertMitjans](#) on Oct 21, 2022



Jaybsoni mentioned this on Oct 24, 2022

[Change eig to eig to support computing the eigenvalues of non-hermitian observables #3204](#)



Jaybsoni closed this as **completed** on Oct 25, 2022

Change `eih` to `eig` to support computing the eigenvalues of non-hermitian observables #3204

< > Code

Merged Jaybsoni merged 8 commits into `master` from `eigvals_bug` on Oct 25, 2022

Conversation 13 Commits 8 Checks 0 Files changed 5

+52 -7



Jaybsoni commented on Oct 24, 2022 · edited by antalszava

Contributor

Context:

When working with linear combinations of operations. If the expressions are not fully simplified, then there may be sub-operators which are not hermitian. When asking for the eigenvalues of such operators, the default pipeline assumes hermiticity and will automatically cast to `float` types (losing information about the complex parts).

We update the pipeline to support complex eigenvalues here.

Related Issues:

[issue 3198](#)

[\[sc-28321\]](#)



made change

6fe06ba



github-actions bot commented on Oct 24, 2022

Contributor

Hello. You may have forgotten to update the changelog!

Please edit [doc/releases/changelog-dev.md](#) with:

- A one-to-two sentence description of the change. You may include a small working example for new features.
- A link back to this PR.
- Your name (or GitHub username) in the contributors section.



Jaybsoni added 3 commits 3 years ago

fixed bug

7223f7d

Merge branch 'master' into eigvals_bug

Verified

548ade8

changelog

158fcfa

Jaybsoni changed the title `[WIP] change-eigh-to-big` `[WIP] change eigh to eig` on Oct 24, 2022

Reviewers

AlbertMitjans ✓
albi3ro ✓
co9olguy ●

Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

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3 participants

